Haoran (Jensen) Yuan

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RESEARCH INTERESTS

My research focuses on **efficient AI**, especially on making **embodied AI generalizable, scalable, and efficient**. I build visuo-tactile world-action models that adapt **pretrained video models** to tactile world modeling **without separate tactile pretraining**, and develop **lossless** parallel drafting methods that speed up LLM inference several times. I am also interested in representation learning and recursive self-improvement (RSI), and I am currently exploring agentic robotics and reinforcement learning for world-action models.

EDUCATION

Doctor of Philosophy in Computer Engineering — Aug. 2026 – May 2028 (Expected)
University of Illinois Urbana-Champaign | Advisor: Prof. Ismini Lourentzou — Champaign, IL

Master of Science in Computer Engineering — Sep. 2024 – May 2026
University of Illinois Urbana-Champaign — Champaign, IL

Bachelor of Science in Electrical Engineering, Minor in Computer Science — Sep. 2020 – Jun. 2024
University of Illinois Urbana-Champaign — Champaign, IL

Bachelor of Engineering in Computer Engineering — Sep. 2020 – Jun. 2024
Zhejiang University — Hangzhou, China

PUBLICATIONS

** equal contribution † project lead ‡ equal advising*

[1] **DexTacWAM: A Visuo-Tactile World-Action Model for Dexterous Manipulation**
Haoran Yuan†, Zekai Wang, Boning Shao, Haoran Lu, Trevor Darrell, Ismini Lourentzou‡, Wei Zhan‡
arXiv, 2026. [project] [arXiv:2609.24976] [code]

[2] **VTAM: Video-Tactile-Action Models for Complex Physical Interaction Beyond VLAs**
Haoran Yuan*†, Weigang Yi*, Zhenyu Zhang*, Wendi Chen*, Yuchen Mo, Jiashi Yin, Xinzhuo Li, Xiangyu Zeng, Chuan Wen, Cewu Lu, Katherine Driggs-Campbell, Ismini Lourentzou
arXiv, 2026. [project] [arXiv:2603.23481]

[3] **JetSpec++: Foresight Parallel Drafting with Self Refinements**
Zhaoxiang Feng*, Lanxiang Hu*, **Haoran Yuan**, Yu-Yang Qian, Jihui Yang, Peng Zhao, Yibo Zhu, Tajana Rosing, Hao Zhang
arXiv, 2026. [arXiv:2610.19792] [code]

[4] **JetSpec: Breaking the Scaling Ceiling of Speculative Decoding with Parallel Tree Drafting**
Lanxiang Hu, Zhaoxiang Feng, Yulun Wu, **Haoran Yuan**, Yujie Zhao, Yu-Yang Qian, Bojun Wang, Peng Zhao, Daxin Jiang, Yibo Zhu, Tajana Rosing, Hao Zhang
arXiv, 2026. [project] [arXiv:2606.18394] [code]

EXPERIENCE

PLAN Lab, University of Illinois Urbana-Champaign — *Urbana, IL*
Graduate Researcher / Ph.D. Student – Advisor: *Prof.* Ismini Lourentzou — Sep. 2025 – Present

VTAM: Video-Tactile-Action Model (*Project Lead*)

- Adapted a pretrained video transformer to touch **without tactile pretraining**, reaching **90%** success on real robots.
- Designed gated cross-attention that cut training loss spikes by **42%** and sped up convergence by **1.7×**.
- Added a force-aware diffusion regularizer to balance tactile–vision attention: **+80%** success over vision-only.

Berkeley AI Research (BAIR), UC Berkeley *Berkeley, CA*
Visiting Researcher – Advisors: *Prof.* Wei Zhan, *Prof.* Ismini Lourentzou, *Prof.* Trevor Darrell Apr. 2026 – Present

DexTacWAM: Dexterous Visuo-Tactile World-Action Model (*Project Lead*)

- Adapted a pretrained video DiT to 10 fingertip tactile sensors via **modality-split flow matching**.
- Designed a finger- and pose-aware tactile compressor: **89.4%** contact recall retained, **3.67**× fewer DiT tokens.
- Achieved **70.6%** avg. success on 6 real-robot tasks vs. **38.0%** for the best baseline; featured by BAIR.

Hao AI Lab, UC San Diego *San Diego, CA*
Research Collaborator – Advisor: *Prof.* Hao Zhang Feb. 2026 – Present

JetSpec++: Foresight Parallel Drafting with Self Refinements (*Core Contributor*)

- Developed a **causal correction for parallel drafting** that conditions each drafted token on its full branch history.
- Outperformed DeepSeek's DSpark at matched size; up to **7.45**× lossless speedup over AR (**1,361 tok/s** on GB200).

JetSpec: Parallel Tree Drafting for Speculative Decoding (*Core Contributor*)

- Co-developed a **causal parallel draft head** that emits a complete speculative tree in one forward pass.
- Achieved up to **9.64**× lossless speedup over AR on Qwen3-8B, beating DDTree (8.78×) and EAGLE-3.

Snowflake *Menlo Park, CA*
Software Engineering Intern May 2025 – Aug. 2025

- Built a **predictive autoscaler** for GS clusters using Gradient Boosting, ARIMA and XGBoost models.
- Achieved **14.3% MAPE** and **92.5%** coverage (95% CI) for 24-hour CPU forecasts, in <5 min per deployment.
- Deployed to adjust instance provisioning, raising average **CPU utilization by >20%** in simulated datacenters.

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, Java, SQL, SystemVerilog.
Frameworks: PyTorch, CUDA, Triton, SGLang, vLLM, Isaac Gym, Isaac Sim, MuJoCo, Flask, SQLAlchemy, OpenCV.
Databases & Tools: MySQL, PostgreSQL, Redis, Docker, AWS, GCP, Git, Linux, Shell.